WIKINSERT: A Multilingual Machine-in-the-Loop Entity Insertion System for Wikipedia

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Abstract

Wikipedia's knowledge network depends on extensive linking between articles, yet entity insertion – adding new contextual links where no anchor text exists – remains a cognitively demanding editorial task. In 60-70% of cases, editors must write new surrounding context before linking a new entity, making this process slow and prone to error at Wikipedia's scale. We present WIKINSERT, a machine-in-the-loop browser plugin-based tool that accelerates entity insertion by creating entity-aware highlighting directly onto live Wikipedia pages. Powered by XLOCEI, our multilingual fine-tuned language model, WIKINSERT computes relevance scores for entity-sentence pairs and delivers them through a lightweight interface that highlights promising insertion points in real-time. The system's three-layer architecture – data pipeline, minimal API, and browser integration – ensures responsive performance while preserving editorial autonomy. Our live demonstration showcases entity search, adaptive thresholds, minimap navigation, and multilingual capabilities across Wikipedia articles in more than 100 languages. Overall, WIKINSERT demonstrates a human-centered AI design, providing visual scaffolding that amplifies rather than replaces human editorial judgment, and exemplifies how AI-powered tools can meaningfully support collaborative knowledge curation at web scale.

1 Introduction

The value of Wikipedia as an encyclopedia stems from its densely interlinked network of articles. Links enable readers to navigate concepts fluidly, yet adding a *new link* in Wikipedia often requires more than just choosing a suitable anchor text. Our previous work shows that in 60-70% of cases, editors must first write new surrounding context before linking a new entity – a cognitively demanding operation known as *entity insertion* (Feith et al., 2024). At Wikipedia's scale, the sheer volume of candidate sentences and the risk of context-breaking edits make manual insertion slow and error-prone.

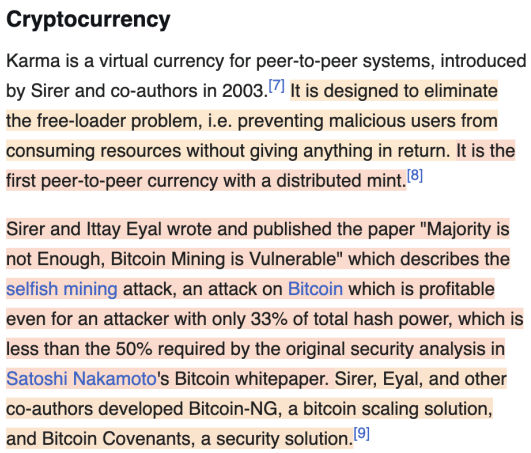


Figure 1: An excerpt from the Wikipedia article "Emin Gün Sirer" showcasing entity-aware highlighting via an overlaid heatmap (redder regions indicate higher relevance, whereas whiter regions indicate lower relevance).

We present **WIKINSERT**, a machine-in-the-loop tool that accelerates entity insertion by *entity-aware highlighting* via a heatmap overlaid directly onto live Wikipedia pages (Fig. 1). Powered by our multilingual XLOCEI framework (Feith et al., 2024), WIKINSERT pre-computes relevance scores for every sentence-entity pair, stores them behind a minimal read-only API, and delivers them through a **lightweight browser extension** that colors sentences in-place (Fig. 2). Editors, therefore, see at a glance where a target entity "fits" and can concentrate effort on rewriting the most promising passages rather than scanning the entire article.

The live demo walks attendees through entity search, heatmap toggling, relevance tool-tips, and threshold adjustment, all on real Wikipedia articles. By bridging *cutting-edge ranking models with a modest and snappy browser extension*, WIKINSERT exemplifies how AI tools can *augment* – rather than replace – human editorial judgement at web scale.

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2 Background and Related Work

This paper positions WIKINSERT at the intersection of two streams of research and tooling: (i) automatic link-maintenance utilities for Wikipedia, and (ii) semantic-highlighting interfaces that help users navigate dense text.

Maintaining Wikipedia's link graph. ADDLINK is a machine-in-the-loop tool that suggests *entity-linking* edits, i.e., inserting hyperlinks where a suitable anchor phrase already exists (Gerlach et al., 2021). Large-scale A/B tests show that newcomers accept 50-60% of AddLink's suggestions, demonstrating the value of lightweight assistance in routine maintenance tasks. However, AddLink does not address the harder *entity insertion* problem, where no anchor phrase is present and new context must be written. WIKINSERT targets this gap by visualizing XLOCEI rankings (Feith et al., 2024) so that editors can decide *where* to add the missing context that will host the new link.

Semantic highlighting for reading and editing. Beyond Wikipedia, the *Semantic Reader* project at Semantic Scholar explores AI-assisted interfaces that overlay contextual cues onto scientific papers. Its SCIM component learns to highlight sentence-level units such as objectives, methods, and findings, enabling rapid skimming without sacrificing comprehension (Fok et al., 2023; Lo et al., 2024). SCIM's success demonstrates that colour-coded overlays, driven by pretrained language models, can reduce cognitive load during complex reading tasks. WIKINSERT adapts this design philosophy to the editorial domain: instead of rhetorical roles, our heatmap encodes model-predicted relevance for entity insertion, giving editors an at-a-glance sense of where their effort is best spent.

Together, ADDLINK and SEMANTIC READER illustrate the promise of AI-assisted micro-interactions for both knowledge curation and deep reading. WIKINSERT combines these ideas–precision link maintenance and semantic heat-mapping into a single tool tailored to the unique demands of Wikipedia editing.

XLOCEI as an enabler. XLOCEI (Feith et al., 2024) is our multilingual fine-tuned language model (LM) that ranks candidate sentences by their suitability for hosting a link to a given target entity. Trained on millions of Wikipedia edits across 105 language editions, XLOCEI achieves state-of-the-art performance – particularly on cases where new text must be written – even outperforming proprietary LLMs such as GPT-4. By visualizing XLOCEI relevance scores, WIKINSERT enables editors to focus effort on the most promising insertion points while keeping all heavy inference offline.

Theoretical Rationale. Our decision to surface XLOCEI 's relevance predictions through a colour-coded sentence heatmap is grounded in *Information Foraging Theory* (IFT), which models readers as involved foragers who follow strong *information scent* to maximise the value of attention invested in text (Pirolli and Card, 1999). In WIKINSERT, sentence-level hues become dynamic scent markers: the deeper the hue, the richer the predicted payoff of inserting the target entity. This enables editors to triage long articles rapidly, concentrating effort on the most promising insertion points and thus reducing cognitive load.

IFT also cautions that scent signals must be both salient and interpretable. We therefore apply the perceptual guidelines of Strobelt et al. (2015), encoding only a single semantic feature per sentence and using three discrete hues (yellow, orange, red) whose contrast produces an immediate pop-out effect without overwhelming neighbouring text.

Finally, the design embodies the Wikimedia Foundation's *human-first AI strategy*, which positions AI as a means of amplifying – rather than replacing – volunteer expertise (Wikimedia, 2025). By restricting AI to transparent cue provision while preserving human editorial authority, this design maintains community autonomy and collaborative control over Wikipedia's knowledge curation.

3 The WIKINSERT System

This section presents the technical architecture and implementation details of WIKINSERT, a browser plugin-based tool for highlighting Wikipedia sentences relevant to entity insertion. The system addresses the dual challenges of computational efficiency and user experience by separating heavy machine learning inference from real-time user interaction through a three-layer architecture (Fig. 2). A *pre-processing layer* first computes sentence-level relevance scores for each target entity; a *backend application programming interface (API) layer* persists and serves those scores; and a lightweight *user interface (UI) layer* renders the heat-map overlay directly inside the live Wikipedia article. WIKINSERT's codebase and resources are available at https://github.com/au-clan/wikinsert.

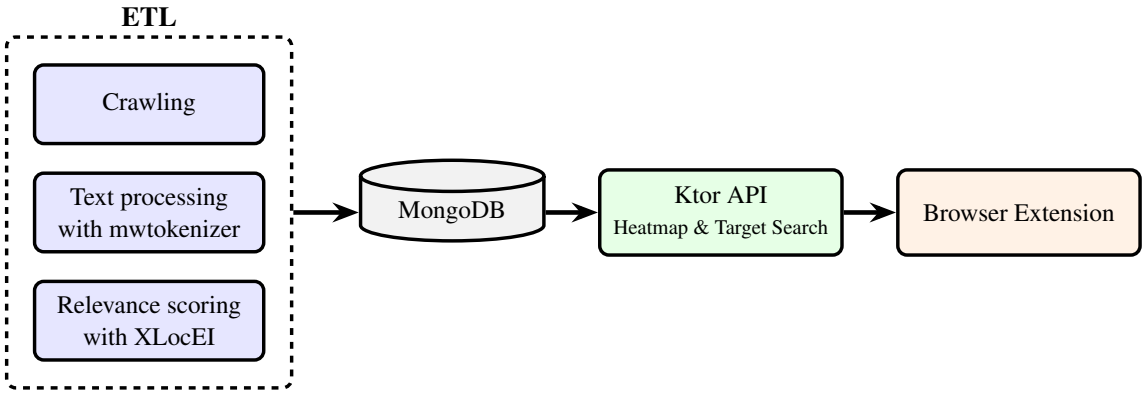


Figure 2: High-level architecture of WIKINSERT. ETL (dashed box) groups three modules: Crawling Wikipedia articles to retrieve content, `mwtokenizer` segments sentences, and XLOCEI assigns entity-sentence relevance scores. The resulting data is persisted in MongoDB. At runtime, a lightweight Ktor API exposes two read-only endpoints – `heatmap retrieval` and `target search` – which supply the WIKINSERT browser extension, allowing it to overlay entity-aware heatmaps on live Wikipedia pages.

3.1 Data Pipeline

Our data pipeline crawls Wikipedia articles, extracts HTML content, and organizes text into sections. This content is then passed through sentence tokenization and XLOCEI relevance scoring to produce data that powers WIKINSERT's user interface.

Sentence splitting with `mwtokenizer`. We decompose each section into sentences using `mwtokenizer`[1], a Wikimedia-maintained sentence splitter built for encyclopedic prose. Unlike general tokenizers trained on newswire, `mwtokenizer` embeds a Wikipedia-specific abbreviation lexicon that correctly handles domain constructs such as "*c. 1700*" and template-generated citations.

For each section string s, we apply language-specific sentence tokenization with abbreviation handling enabled and maintain a running character pointer `offset`. For the i-th sentence we record $(idx_i, start_i, end_i, sent_i)$ where $start_i = \texttt{offset}$ and $end_i = \texttt{offset} + |sent_i|$. This global-offset scheme ensures sentences can be relocated unambiguously in the original HTML DOM.

The tokenizer is extremely fast (2 ms for a 5 kB article) and robust across 280+ Wikipedia language editions. Sentence records are grouped by section title and persisted in a MongoDB database, providing a canonical index for downstream components.

Relevance Scoring with XLOCEI With sentences extracted, we compute their relevance to target entities using the XLOCEI model (Feith et al., 2024). For each candidate sentence, we construct model input by concatenating: (i) the target entity article title, (ii) the target's lead section, (iii) known entity mentions, and (iv) the candidate sentence with local context (five preceding and five following sentences within the same section). This ten-sentence context window provides topical context while maintaining computational efficiency.

The model outputs numerical relevance scores for each sentence-target pair. These scores are as well persisted in our MongoDB, indexed by source and target article IDs. Each record contains the sentence identifier, start/end character offsets, and relevance score. By storing global character offsets, we enable accurate alignment between model outputs and in-browser highlighting functionality.

3.2 Backend Architecture

The backend layer consists of a MongoDB database for storing preprocessed sentence records and relevance scores, and a Ktor-based REST API that serves the browser extension. The extension communicates with the backend through just two endpoints: one that returns the heatmap for a given `(article, target)` pair and one that provides entity search. Restricting the interface to read-only operations improves robustness under flaky network conditions and ensured – in formative usability tests – that participants never encountered failed requests or cryptic error states.

3.3 Browser Integration

The UI layer is implemented as a browser extension to maintain editorial workflow continuity within the familiar MediaWiki environment. This approach eliminates context switching and data synchronization issues while ensuring that highlights appear precisely where subsequent edits will be made.

[1] https://github.com/wikimedia/mediawiki-extensions-NLPTools

Score normalisation and threshold clustering. Raw xLocEI relevance scores exhibit significant variance across articles and targets, necessitating normalisation to a common $[0, 1]$ scale before clustering into discrete intensity bins. We implement two complementary normalisation pipelines: (i) *z-score + sigmoid* normalisation that recentres scores around the article mean and applies smooth compression, followed by Jenks natural breaks clustering to minimize intra-class variance; and (ii) *empirical-CDF* normalisation that maps scores to rank percentiles with fixed interval cuts $(0.40, 0.55, 0.65, 0.75, 0.85)$, ensuring uniform coverage across content types. Both methods provide distinct editorial perspectives: adaptive clustering emphasizes score differentials within articles, while rank-based clustering ensures predictable workload distribution.

Heatmap color generation. Normalised relevance scores $r \in [0, 1]$ are converted to perceptually meaningful hues through a two-segment RGB mapping. For scores $0 \leq r \leq 0.5$, red and green channels remain at maximum (255) while blue fades linearly from 255 to 0, creating a white-to-yellow transition. For higher scores $0.5 < r \leq 1$, red remains saturated while green reduces linearly and blue increases, producing a yellow-to-orange-to-red progression that peaks at the WCAG-AAA compliant tone `#FF5E5E`. An opacity term $\alpha(r) = \frac{r+0.3}{4}$ maintains text readability while enhancing visual salience. Sentences below the 0.40 threshold are suppressed to reduce visual clutter and focus attention on actionable insertion candidates.

Performance optimization. To preserve the fluid interaction pace expected in Wikipedia editing workflows, WikInsert maintains sub-100ms response times through architectural optimization. Heavy computation – sentence segmentation and xLocEI inference – is executed in our ETL pipeline, with the browser receiving only compact JSON arrays of sentence offsets and scores. The content script maps these offsets to DOM nodes using native APIs, while tooltip display and minimap scrolling employ event-delegated listeners with session caching to eliminate network requests for subsequent interactions. These optimizations collectively sustain responsive user experience while coexisting with Wikipedia's possibly large pages.

4 User evaluation

To evaluate Wikinsert, we conducted a preliminary think-aloud study with 4 university students representing novice Wikipedia editors. Participants completed entity insertion tasks across 4 different Wikipedia articles using Wikinsert's entity-aware highlighting functionality. Since the system lacked an in-place editor, participants used an "edit sheet" to record their proposed insertions. Each session was recorded with screen capture and audio commentary. For 3 of the 4 articles, at least 3 users found the system to be useful. For the fourth article, an entire section was highlighted, making it less helpful as it obscured fine-grained relevance distinctions. The study revealed key usability issues, including problems with the unlabeled intensity slider and the need for clearer visual feedback. These findings informed concrete design improvements in the second iteration: replacing the slider with semantic presets ("Very Low" to "Very High"), adding hover tooltips with relevance scores, and implementing automatic scrolling to high-relevance sentences. The study validated Wikinsert's core design philosophy of augmenting rather than replacing editorial judgment.

Leveraging our strong collaboration with the Wikimedia Foundation, we will evaluate WikInsert on Wikipedia as a successor to AddLink and the novel task of de-orphanizing Wikipedia articles (Arora et al., 2024).

5 Demonstration

The live session guides the audience through a complete entity insertion workflow on a real Wikipedia article. The tour is organised around WikInsert 's four locality layers:

1. the **toolbar popup**, where a target entity is searched, selected, or cleared;
2. the **interaction menu**, which lets editors adjust highlight threshold and scaling on the fly;
3. the **document minimap**, providing global context and click-to-scroll navigation; and
4. the **in-place tooltips** that disclose sentence-level relevance scores on hover.

All interactions are executed on a laptop running the extension against the live Wikipedia site and streamed in real time via screencast; a public staging build allows participants to reproduce every step on their own machines.

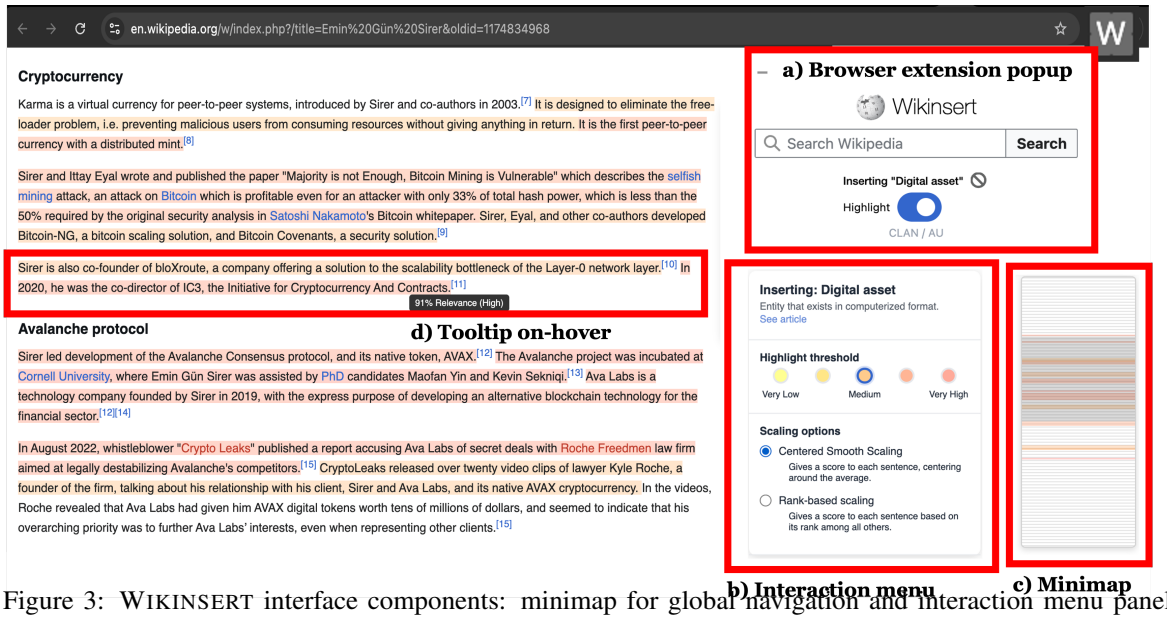


Figure 3: WIKINSERT interface components: minimap for global navigation and interaction menu panel for threshold adjustment. (Layout shifted for the sake of demonstration)

5.1 Extension Popup

This streamlined toolbar popup acts as the central control interface for editors. A single type-ahead field streams candidate entities – complete with thumbnails and short descriptions – as the user types, allowing rapid, search of our namespace. Selecting an entity instantly commits the `(source, target)` pair to session storage and requests the corresponding heatmap. The same popup also provides session-level controls: a toggle switch shows or hides the overlay without reloading the page; the "clear" button removes the current target and removes the heatmap; and, at any time, the search box can be reused to pick a new entity. By collocating search, activation, and reset in one lightweight dialog, the design keeps cognitive load minimal while ensuring entity-insertion assistance is always just one click away from the standard Wikipedia workflow.

5.2 Interaction Menu: Threshold & Scale Control

Whenever highlighting is active, a slim, docked info card stays pinned to the right edge of the viewport (Figure 3 (b). The header reminds the editor which target entity is in focus, while two control blocks let them shape the heat-map in-place.

Threshold. Five radio buttons – *Very Low* through *Very High* – trade recall for precision: the lowest setting paints a broad scaffold of faint cues across the article, whereas the highest isolates only the top few sentences, useful when the editor already knows roughly where to insert.

Scaling. A second toggle chooses between *Smooth* (Z-score + sigmoid) and *Rank-based* (empirical CDF) normalisation, giving alternative "flavours" of the same relevance landscape. Editors can thus switch from an absolute view, centred on the article's mean score, to a relative view that highlights only the best-ranked sentences, regardless of score variance.

This positioning maintains editor orientation and prevents the spatial disorientation documented in preliminary testing. Additionally, the interface enables seamless experimentation with different thresholds and scalings – each adjustment re-renders locally using the data from the session storage without requiring network communication.

5.3 Minimap: Global Context & Rapid Navigation

A slim "overview strip" sits just beneath the interaction menu, condensing the whole article into lane of colour bars (Figure 3 (c)). Each bar mirrors a highlighted sentence: its vertical position and height correspond to the sentence's offset and length in the document, while its hue matches the main heatmap. A translucent grey band indicates the current viewport position, providing continuous spatial orientation within the document context. Beyond passive orientation, the minimap acts as a spatial index: clicking any point triggers a smooth scroll that lands the selected region midddle of the viewport, letting editors jump between hotspots in one gesture. The component is built from lightweight, absolutely positioned `divs`; thus it repaints instantly

when thresholds or scaling schemes change, rendering almost instantaneously.

5.4 In-place Relevance Tooltips.

When the cursor hovers over any coloured sentence, a compact tooltip appears beside the pointer showing two facts drawn from the loaded heatmap: (i) the normalized score rounded to an integer percentage (e.g. "78 %") and (ii) a verbal descriptor (*Very Low - Very High*) derived from the current set of dynamic categories. Because all scores are already held in session memory, the lookup is latency-free and the tooltip fades in within one animation frame, disappearing as soon as the cursor leaves. This explicit numerical read out grounds the qualitative heatmap in a quantitative signal, letting editors decide whether a "Medium" location is worth rewriting or to keep searching for more higher scoring candidates.

6 Use-Case Discussion

Expert aid versus novice onboarding. WIKINSERT can be positioned either as a precision aid for veteran editors or as a scaffolding device for newcomers. Seasoned contributors often arrive with a clear hypothesis – "*this page ought to mention X*" – and merely need rapid visual confirmation of where the missing link belongs; the heat-map therefore acts as a time-saver without intruding on their established workflow. Conversely, newcomers face a larger search space and are still learning Wikipedia's style conventions. Here the same overlay doubles as an instructional lens: by comparing high-scoring sentences with low-scoring ones, novices implicitly absorb what "good" contextual framing looks like.

Capturing editor-supplied context. Entity names seldom come in isolation; editors frequently know *why* the link matters. That rationale is lost when inferring only with the target article. A natural extension is a dual-input interface that asks not just "*Which entity?*" but also "*What aspect is relevant?*" a free-text fragment could be concatenated to the model prompt or distilled into structured cues (topic keywords, section hints), enabling more accurate scoring and post-hoc explanations. Such mixed-initiative design would align with Wikipedia's human-first AI ethos: editors contribute domain insight, the system amplifies it, and both parties share accountability for the final edit.

7 Discussion and Concluding Insights

WIKINSERT demonstrates how AI-powered tools can meaningfully augment human editorial judgment in Wikipedia's collaborative knowledge curation process. By translating XLOCEI's relevance predictions into an intuitive visual interface, the system addresses the cognitively demanding task of entity insertion while preserving editorial agency and expertise.

The three-layer architecture – offline computation, lightweight API, and browser-based visualization – proves that sophisticated AI assistance can be delivered through minimal, responsive interfaces that integrate seamlessly into established workflows. Rather than disrupting Wikipedia's editing culture, WIKINSERT amplifies existing editorial practices by providing visual scaffolding that helps editors navigate complex textual landscapes more efficiently.

Our demonstration reveals the potential for AI-assisted tools that respect human expertise while reducing cognitive burden. The heatmap interface serves as suggestive guidance rather than prescriptive automation, enabling editors to develop effective mental models and adapt the tool to their individual search strategies. This approach embodies the Wikimedia Foundation's human-first AI philosophy, where technology enhances rather than replaces collaborative intelligence.

By tackling the entity insertion problem – where 60–70% of new links require writing contextual text – WIKINSERT directly supports Wikipedia's mission to bridge knowledge gaps, addressing the critical issue of orphan articles that remain invisible to readers. The multilingual capabilities of the underlying XLOCEI framework extend this impact to underrepresented languages where such gaps are most pronounced.

The success of WIKINSERT ultimately lies in its recognition that the most valuable AI-assisted tools are those that amplify human capabilities while preserving the collaborative, accuracy-driven culture that makes Wikipedia a trusted global resource. Through a human-centered AI design, we demonstrate a path forward for human-AI collaboration that enhances editorial efficiency without compromising editorial autonomy and preserving Wikipedia's AI strategy of "putting humans first".

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A First iteration of WIKINSERT

B Participant Script for Think-Aloud Study

This appendix represents the planned structure of each think aloud study from the facilitator's side. The task states the source article and the underlined part is the target entity.

1. Welcome & Framing

> Thanks for helping us today!
>
> Imagine you're a regular Wikipedia editor: you have reliable sources in front of you and know what needs to be added, but you still have to decide where in the article it belongs.
>
> The tool we're testing, Wikinsert, overlays a colour heat-map on each sentence to show how semantically relevant it is to the fact you want to add. Your job is to use that heat-map to choose the best insertion point and draft the sentence.
>
> You may re-word the fact in your own style, but its meaning must stay intact, specifically you have to link the target entity in some way. Speak your thoughts out loud as you go; there are no wrong answers.

2. Task: Porphine (Chemistry)

> A short note in the Journal of Heterocyclic Chemistry clarifies that $C_{20}H_{14}N_4$ is the empirical formula of Porphine.
>
> Please add this fact (in your own words) to the Porphine article.

3. Task: Barbuda (Political Science)

> You just read a well-known civil and political science handbook that states:
>
> *The political system of Antigua and Barbuda is a representative democracy.*
>
> Please weave that information into the Barbuda article.

4. Task: Mizone (Sports Nutrition)

> An independent sports-science study (2008) reports:
>
> *In a test with cyclists and triathletes, Mizone was absorbed up to 16 % faster than Powerade or other popular sports drinks.*
>
> Please add this research finding to the Mizone article.

5. Task: Elseworld (Comics Publishing)

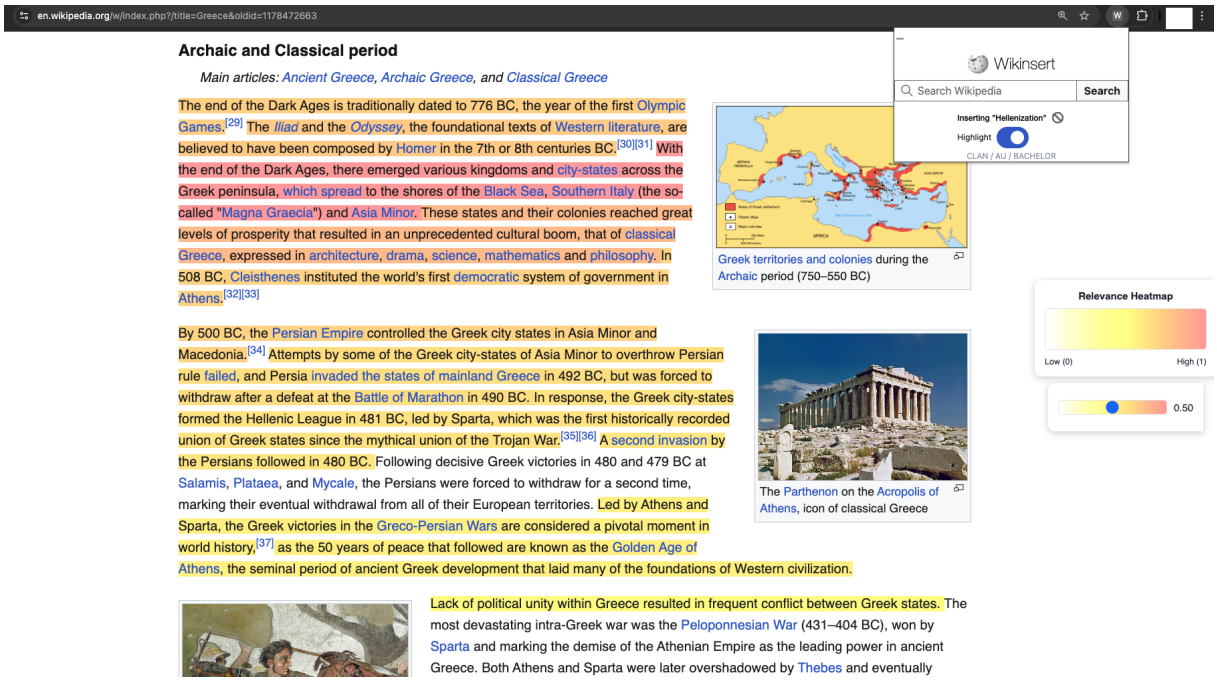


Figure 4: First iteration of the Wikinsert extension overlaying an entity-aware heatmap on a Wikipedia article.

> News from New York Comic-Con (12 Oct 2023) announced:
>
> *Andy Diggle will write Gotham by Gaslight: The Kryptonian Age as part of DC's 2024 Elseworlds relaunch.*
>
> Please incorporate this announcement into the Elseworld article.

6. Wrap-Up Questions

1. What made a sentence feel like the right place?
2. Was there ever a time the heatmap misled you?
3. If you could change one thing about Wikinsert, what would it be?

C Multilingual Interface Screenshots

This appendix presents screenshots of Wikinsert's interface across different Wikipedia language editions, demonstrating the system's multilingual capabilities and cross-lingual entity insertion functionality.

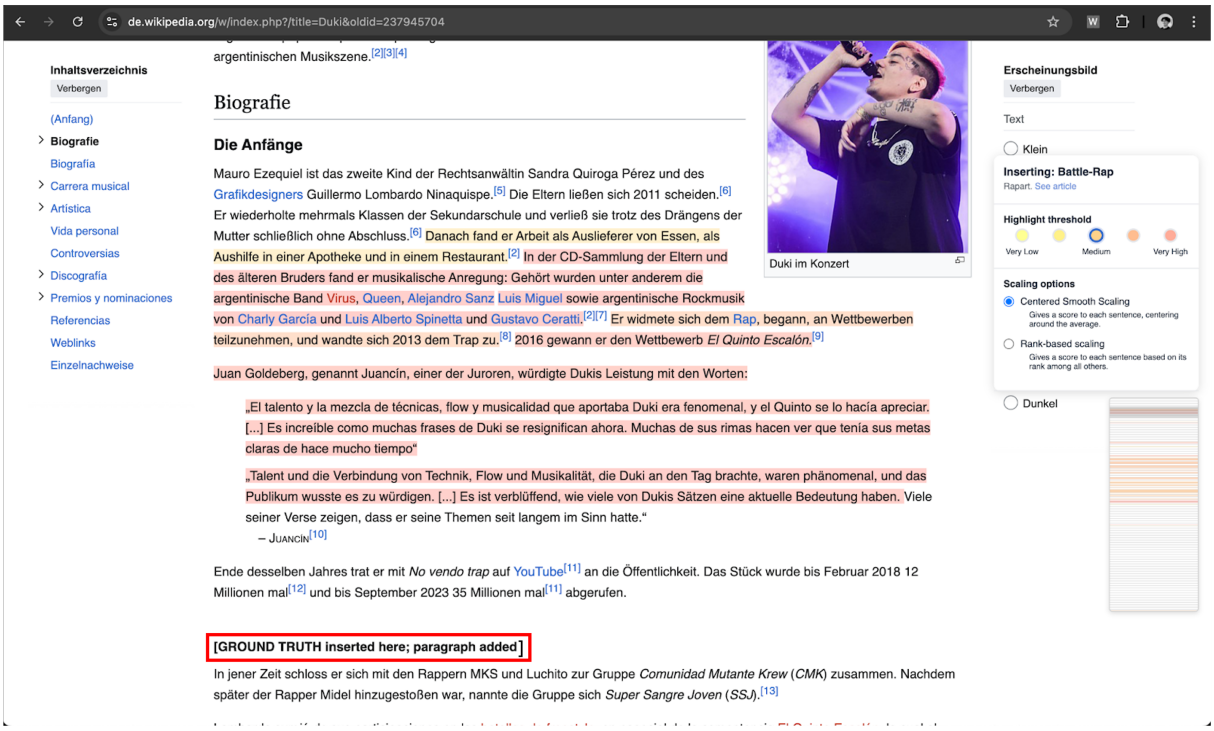


Figure 5: Inserting the target article "Battle-Rap" to source article "Duki", in the German Wikipedia. The target was inserted in the next revision by adding a new paragraph.

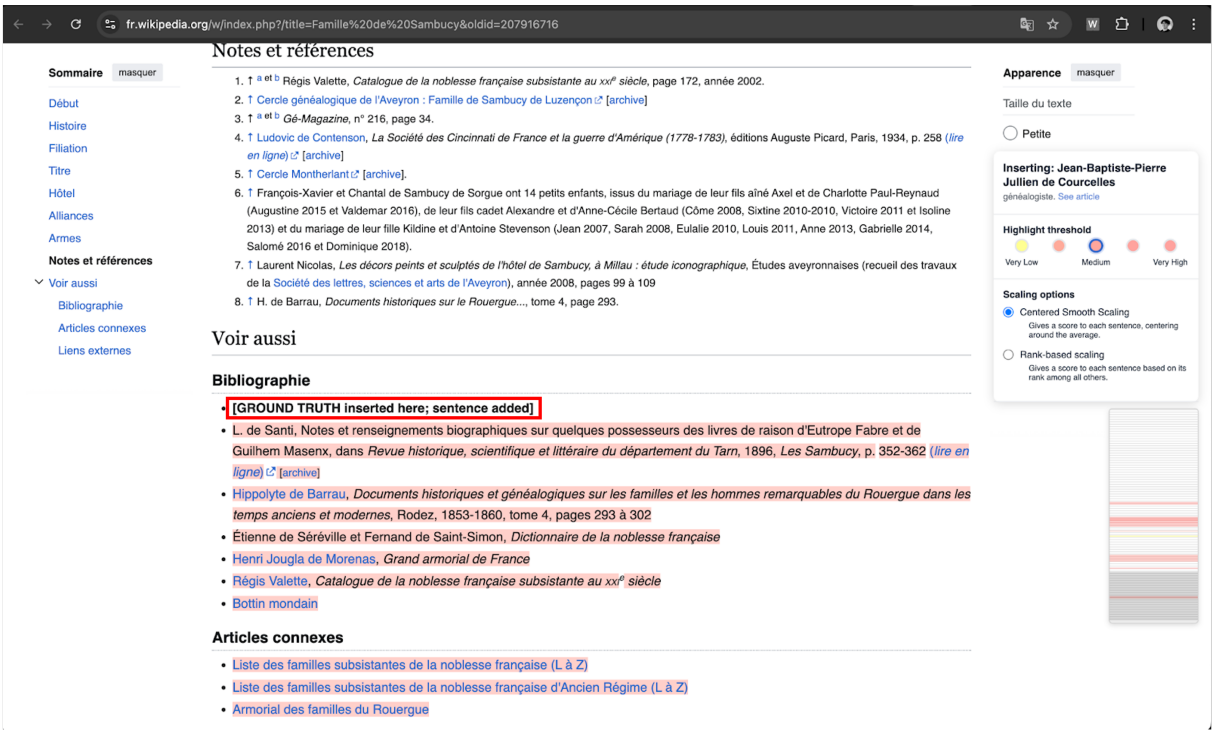


Figure 6: Inserting the target article "Jean-Baptiste-Pierre Jullien de Courcelles" to source article "Famille de Sambucy", in the French Wikipedia. The target was inserted in the next revision by adding a new sentence.

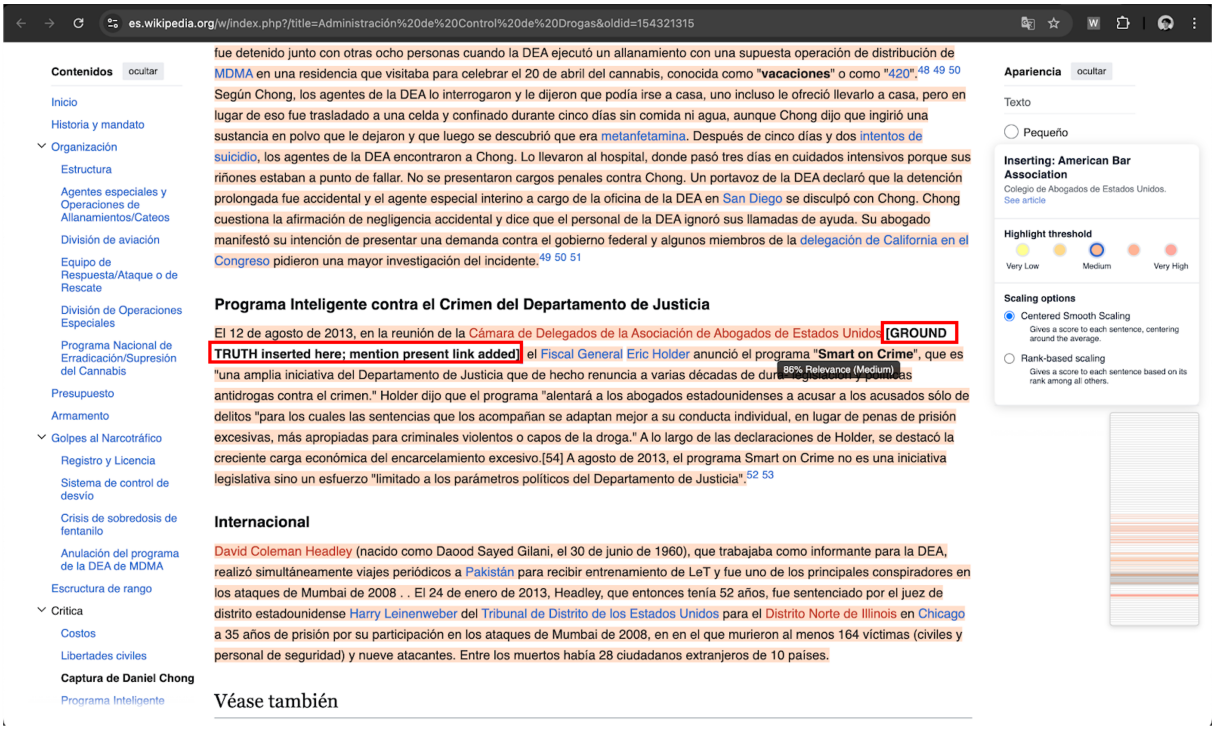


Figure 7: Inserting the target article "American Bar Association" from the English Wikipedia to source article "Administración de Control de Drogas" in Spanish Wikipedia (a cross-lingual insertion). The target was inserted in the next revision by only linking as the mention was already present.



Figure 8: Inserting the target article "乃村健次" (Kenji Nomura) to source article "EDENS ZERO角色列表", (EDENS ZERO character list) in the Chinese Wikipedia. The target was inserted in the next revision by adding the mention with no additional surrounding text.



Figure 9: Inserting the target article "Loan" to source article "Grand Continental Hotel" , in the Arabic Wikipedia. The target was inserted in the next revision by only linking as the mention was already present.

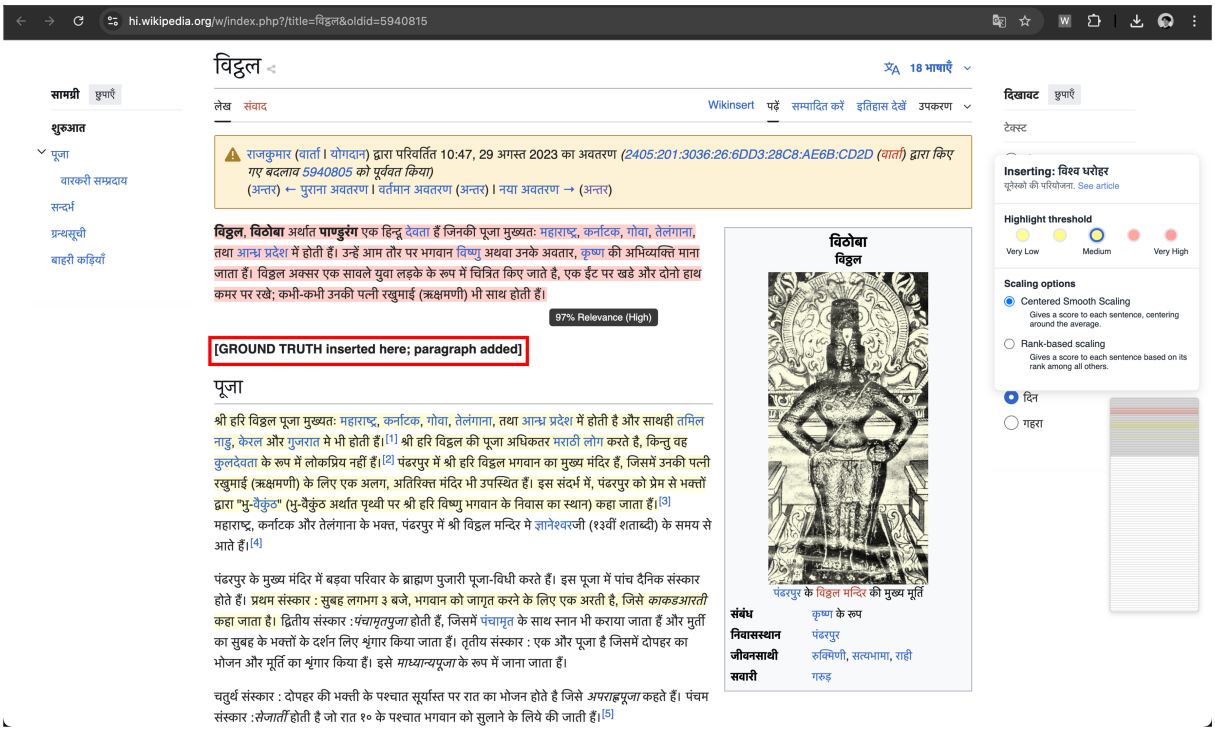


Figure 10: Inserting the target article "" to source article "", in the Hindi Wikipedia. The target was inserted in the next revision by adding a paragraph.